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FARMER PARTICIPATORY CROP IMPROVEMENT. II. PARTICIPATORY VARIETAL SELECTION, A CASE STUDY IN INDIA

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SUMMARY

Farmer participatory varietal selection (PVS) was used to identify farmer-acceptable cultivars of rice and chickpea. Farmers' requirements in new crop cultivars (varieties) were determined, a search was carried out for released and non-released cultivars that matched these needs, and they were tested in farmer-managed, participatory trials. Farmer-acceptable cultivars were found amongst released material, but not among the recommended material for the area. Lack of adoption is, therefore, because resource-poor farmers have not been recommended or exposed to the most appropriate cultivars under the existing system of varietal identification and popularization. Adoption rates of cultivars would be improved by increased farmer participation, the systematic testing in zonal trials of locally popular cultivars to define their domains properly, a more liberal release system, and a more open system of providing seeds of new cultivars to farmers.

INTRODUCTION

The Krishak Bharati Cooperative Ltd (KRIBHCO) Indo British Rainfed Farming Project (KRIBP), is a participatory development project financed by the UK Overseas Development Administration (ODA) and the Government of India. Initial surveys at the project planning stage had shown that, in common with many marginal environments in India, the uptake of improved cultivars was extremely low. At the inception of the project a programme of farmer participatory varietal selection (PVS) was planned. The methods of PVS employed (Witcombe *et al.*, 1996) were designed to identify and overcome the constraints that caused farmers to continue to grow landraces. In the first three years of the project PVS programmes were conducted with several crops, including maize (*Zea mays* L.), chickpea (*Cicer arietinum* L.), black gram (*Phaseolus mungo* L. = *Vigna mungo* (L.) Hepper), and pigeonpea (*Cajanus cajan* L.). The research on upland rice and chickpea is described in detail. In the light of the results, alternative

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strategies are discussed for the breeding, identification, release and popularization of cultivars that are better at meeting the needs of resource-poor farmers.

MATERIALS AND METHODS

Participatory rural appraisal

The project area consists of the adjoining districts of Panchmahals (Gujarat), Jhabua (Madhya Pradesh) and Banswara (Rajasthan). Annual rainfall is 700–1100 mm and the infertile, shallow soils are on sloping land. The farmers have few resources with average land holdings of less than 1 ha. Maize is the most important crop in the rainy season (*kharif*) but upland, direct-drilled rice is also widely grown. Most farmers cultivate chickpea in the post-rainy season (*rabi*), utilizing residual soil moisture. Participatory rural appraisals (PRAs) were conducted on the major characteristics of the local cultivars. This was done in village meetings by focused group discussions (FGDs). A number of other PRA techniques were employed such as 'matrix ranking' where local landraces were assessed by ranking them for multiple traits. The FGDs were conducted by village-based project staff (called community organizers). The FGDs were with groups of 6 to 20 farmers, both men and women, and usually all the socio-economic classes perceived by the community were represented adequately. Crop-focused PRAs were conducted in the villages of: Chatra Khunta (Rajasthan) in August 1992; Kompura (Rajasthan) and Jaliyapada (Gujarat) in October 1992; Bor (Gujarat) and Sarjumi (Gujarat) in November 1992; and Palasiyapada and Naganwat Choti (both in Madhya Pradesh) in December 1992.

Search and procurement

A search was carried out for rice cultivars released at the national and state level that matched, or were superior to, the local landraces for traits identified as important in the PRAs. Once cultivars were identified in the search, attempts were made to procure seed from breeders and commercial sources.

Particular efforts were made to find cultivars of rice released at the state level outside the project area. Published information was examined but proved inadequate. Personal visits were made to obtain recent, accurate information. Visits were made to the Indian Council for Agricultural Research Institutes, State Agricultural Universities (SAUs) and seed producers. In the three project states, visits were made to the agricultural universities, and in Gujarat and Madhya Pradesh to district-level officials in the State Departments of Agriculture, and to State Seed Corporations. State Departments of Agriculture provided information only on released cultivars recommended in the package of practices by the Krishi Vigyan Kendras (farm science centres). Outside these states, visits were made to the Indian Agricultural Research Institute, New Delhi, the Directorate of Rice Research at Hyderabad, Andhra Pradesh, the Central Rice Research Institute at Cuttack, Orissa, and the Narendra Dev Agricultural University at Faizabad, Uttar Pradesh.

Tunwar and Singh (1985) provided useful information for rice cultivars released up to 1985. Thereafter, information was more difficult to obtain because varietal characteristics are not described in the *National Catalogue* (Ministry of Agriculture, Government of India, 1993) and the *Proceedings of the Central Variety Release Committee*, which does contain such varietal information, are not widely circulated. In a second visit to the Central Rice Research Institute (CRRI) at Cuttack, in 1994, a more detailed compilation of rice cultivars was obtained and this included releases up to 1991 (CRRI, 1992).

The search for chickpea cultivars was made through the chickpea breeders at the International Crops Research Institute for the Semi Arid Tropics (ICRISAT). They were able to identify and provide seed of four phenotypically diverse cultivars that were likely to be adapted to the project area.

Conduct of the farmer-managed trials

Varieties were tested with farmers in farmer managed participatory research (FAMPAR) trials. These were divided into introductory trials and adaptive trials. In the introductory trials small quantities of seed were given to farmers, but in the adaptive trials seed was sold at commercial rates to avoid overestimating cultivar acceptability. The trials were the simplest possible: each participating farmer was assigned a single variety at random which was grown alongside the local variety in the same field without any changes in management. Each farmer was given two 1-kg bags of rice seed so that a second bag was available for resowing if required. Sometimes, the farmer grew a row of a different crop between the two cultivars to mark the plots. The average plot size with 1 kg seed was 100 m^2. For chickpea, farmers received only one 1 kg bag of seed because replanting is not possible on unirrigated land in the *rabi* season.

Trial details are shown in Table 1. The 1993 trials in the *kharif* (rainy season) were planned with 150 participating farmers, but successful trials were conducted by only 128 farmers because not all of them planted the seed. The lowest number of replicates for a cultivar within a village was 3, instead of the planned 5.

Evaluation of the trials

Farm walks were organized for the participating farmers to visit each other's plots so that all the cultivars could be compared in the discussions. In the introductory trials, data were collected before and after harvest by means of focused group discussions (FGDs) with all participating farmers on aspects of the crop including taste, market value, threshing characteristics and storability. For a few selected pre-harvest and yield traits, the test entries were scored as better, the same or worse than the local landrace. The focused group discussions were followed by detailed household level questionnaires (HLQs) completed by individual households. These were used to assess the household's perception of the variety in terms of post-harvest traits such as cooking quality and the market value of the grain. In 1994, in the adaptive trials, yields of Kalinga III and the local cultivar were measured in four villages in a total of 58 paired comparisons.

Table 1. Introductory and adaptive farmer managed participatory research (FAMPAR) trials with rice and chickpea, 1992–94

Crop	Season	Level	Test cultivars	Number of villages	Total number of farmers
Rice†	1992 rainy	Introductory	GR-3 JR-75 Sathi-34-36 GR-5 Poorva	2	45
Rice‡	1993 rainy	Introductory	Kalinga III Jaldi Dhan-1 GR-3 Sathi-34-36 Jaldi Dhan-3	6	128
Rice	1994 rainy	Adaptive	Kalinga III Sathi-34-36	12	—
Chickpea§	1992-93 post-rainy	Introductory	ICCC 4 ICCV 37 ICCV 2 ICCV 10 ICCV 88202	6	120

Experimental design: †4 farmers per variety in 1st village, 5 farmers per variety in 2nd village, 5 varieties per village; ‡5 farmers per variety in each village, 5 varieties per village; §4 farmers per variety in each village, 5 varieties per village.

The area of each plot was estimated by researchers and, because farmers measured the amount they harvested reliably, the total yield from the plot determined from farmers' measured yield.

In chickpea trials in 1992–93, quantitative data on various traits, including yield and maturity, were collected for one plot of each cultivar in each of the six villages. In 1993–94 there were adaptive trials, but the data are not presented. In 1995, a follow-up survey was made to examine resowing rates in 1993–94 and whether farmers gave or sold seed to others.

Statistical analyses

Farmers scored the performance of Kalinga III rice in three classes: better, the same or worse than the local. The percentage data were transformed to arcsines and analysed by a single classification analysis of variance with villages as replicate blocks within a class. The between-classes mean square was tested against the pooled within-class mean squares with 15 degrees of freedom. Standard errors for the class means were calculated from the within-class variance using untransformed data. When differences were significant they were further partitioned into two orthogonal comparisons with one degree of freedom.

The yield data were compared with a one-tailed paired Students *t*-test. The coefficients of determination (r^2) were calculated for the relationship between the yield of Kalinga III and the local landraces across farmers within each village. A risk aversion analysis for Kalinga III with the local landraces was done using the mean-standard deviation analysis of Binswanger and Barah (1980) for a single year's testing. The method for a single-year analysis is described in Witcombe (1988).

RESULTS AND DISCUSSION

Participatory rural appraisals (PRA)

Prior to the current experimental programme, there was no adoption of improved cultivars in any crop other than cotton. The PRAs revealed that the improved cultivars available on the market did not meet the farmers' needs. They had one or more undesirable traits, usually they took too long to mature, and cereals often yielded too little straw. Many farmers believed that improved cultivars required additional inputs, such as fertilizer or irrigation to perform well enough to provide benefit, and either they had no access to these inputs or they could not afford or risk their purchase. The perception that improved cultivars required higher inputs was due in part to the traditional 'recommended package of practices' approach of scientists and extension workers.

Seven direct-drilled rice landraces, and three transplanted rice landraces were identified in the PRAs. However, this was a conservative estimate, because landraces with almost identical names were assumed to be the same and only a sample of farmers was surveyed. Direct-drilled rice was much more important than transplanted rice. All of the direct-drilled landraces were early maturing and were of tall or medium plant height. Only one local cultivar of chickpea, Dahod yellow, was grown.

The PRAs in the village proved to be a quick and effective method of identifying and characterizing the landraces which farmers grew. The results of subsequent trials showed that they were an essential first step in choosing which cultivars should be tested with farmers. Nonetheless, there are limitations as to what a PRA can reveal and the merits of collecting and growing out landraces are discussed in detail by Witcombe and colleagues (1996).

Search and procurement of rice cultivars

In the extensive search, described in the Materials and Methods section, 15 suitable, released cultivars of rice from six sources were identified. Seed from three of the cultivars was unobtainable, and testing of another three cultivars was delayed until seed was obtained for the 1995 season. Four cultivars, GR-3, GR-5, JR-75, and Poorva, were included because they were released officially in the project area and were recommended for dryland farming. However, they did not match the characteristics of the farmers' cultivars because they were dwarf or semi-dwarf cultivars.

The search process proved to be more difficult than anticipated. Up-to-date published information on which cultivars were released, their major characteristics, and the agro-ecological system to which they were best adapted was difficult to find.

Copies of the useful publications that do exist (for example, Roy, 1989) are not widely available. Access to unpublished information was essential and required personal visits involving considerable time and cost. Many organizations wishing to undertake a participatory varietal selection (PVS) programme would not have resources for the visits required to identify the most promising cultivars of rice and chickpea. Indeed, any organization operating in the project area that restricted the search for cultivars to those recommended by the official system would have failed to identify a suitably adapted cultivar. Information and seed need to be readily available for non-governmental organizations, or any other organization, wanting to use the PVS methodology for providing a range of varietal choices to farmers. The latest information needs to be published and accessible.

The lack of a centralized seed supply mechanism meant that many letters and personal visits to seed producing agencies were required. A better mechanism is needed in India for making relatively small quantities of seed available for participatory research.

At an international level, few national agricultural research systems have produced the diversity of cultivars available in India. Most have relied heavily on the introduction of international nurseries of advanced or segregating lines from the international agricultural research centres. Instead, national agricultural research systems need to introduce cultivars which have been released by other national programmes that match local cultivars in important characteristics. Unfortunately, free exchange of cultivars is often restricted between countries unnecessarily. One reason may be the belief that intellectual property rights are being defended, but often there are no plant breeders' rights in the countries concerned.

Comparison of rice cultivars, rainy season 1992

The rainy-season trials were conducted in collaboration with a non-governmental organization, the Sadguru Water and Development Foundation at Dahod in Gujarat. The trials were in two of their project villages, Gangardi and Jhari Bujarg, in Gujarat. The cultivars were all ones which were released for Madhya Pradesh and Gujarat and could be obtained easily there. Rainfall was below average in 1992, and in farmers' fields with low-input conditions all except Sathi-34-36, produced little or no grain. Farmers liked Sathi-34-36, an old variety released in 1955, and it was the only tall cultivar tested. It is necessary to match test cultivars with the local material in terms of a character, such as plant height, perceived by farmers to be important.

Comparison of rice cultivars, rainy season 1993

In 1993 farmers' perceptions of the five test cultivars, the previously identified Sathi-34-36 and four new cultivars, were assessed by focused group discussions and household level questionnaires.

Nearly all farmers considered that Kalinga III yielded more than the local landrace and about half considered that Sathi-34-36 yielded more than the local

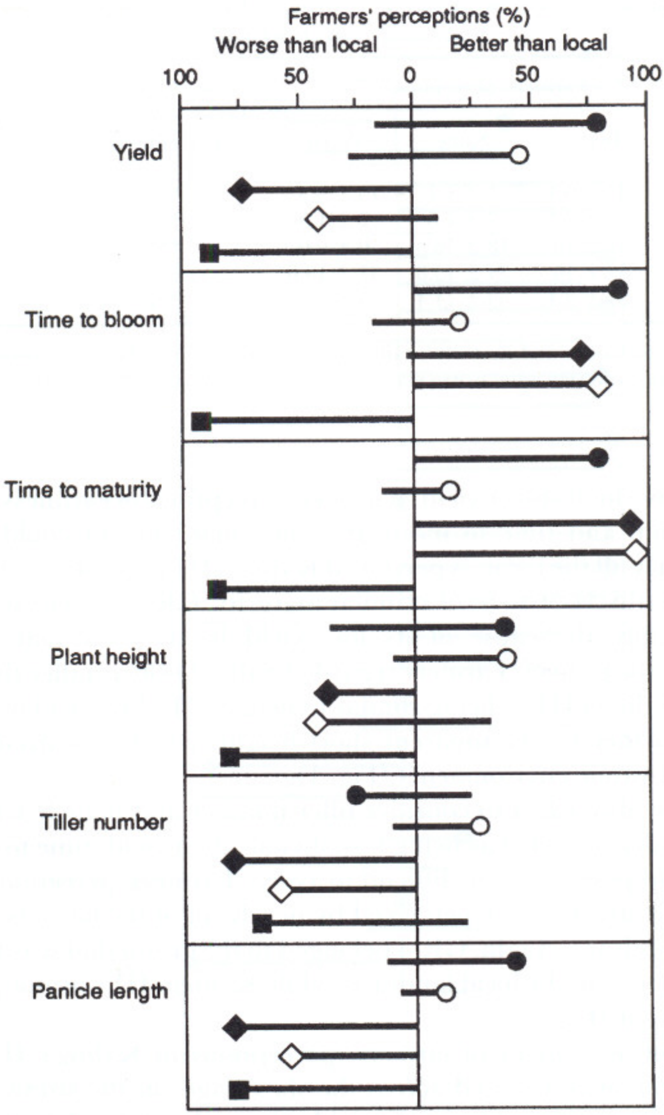

Fig. 1. Farmers' perceptions of five rice cultivars evaluated in the 1993 rainy season. Farmers' perceptions as to whether the test cultivars were better or worse than the local landrace are indicated by lines. The shorter the line for any cultivar, the more farmers responded that the cultivar was the same as the local landrace —● Kalinga III, —○ Sathi-34-36, —◆ Jaldi Dhan-1, —◇ Jaldi Dhan-3, —■ GR-3.

landrace (Fig. 1). GR-3, the released and recommended cultivar, was perceived by farmers to be the lowest-yielding entry in the trial by far and its very low yield was equally obvious to the project team members. Farmers perceived Jaldi Dhan-3 and Jaldi Dhan-1 to be better than GR-3, but markedly inferior to the local landraces.

From the participatory rural appraisals and focused group discussions it was known that farmers preferred earlier-maturing rice genotypes and tall genotypes

Table 2. *Statistical analysis of farmers' perception (%) as to whether Kalinga III was better, worse or the same as the local landrace, rainy season 1993*

Trait	Farmers' perceptions (%)			Significance between farmers' perceptions	Significance of comparison: more *versus* same + worse‡
	Better	Same	Worse		
Grain yield	83 ± 9.4	1 ± 1.1	15 ± 9.3	***	***
Time to bloom	95 ± 2.1	3 ± 1.8	2 ± 1.9	***	***
Maturity	82 ± 16.5	18 ± 16.5	0 ± 0.0	***	***
Height	39 ± 16.7	24 ± 15.8	37 ± 19.0	NS	—
Tiller Number	23 ± 8.4	53 ± 14.4	25 ± 15.5	NS	—

†Earliness for time to bloom and maturity, tallness and high tiller number were better; ‡Remaining orthogonal comparison of same *versus* worse was always non-significant; ***$p < 0.001$.

because they have high stover yield. Farmers' perceptions of earliness were scored as time to bloom and time to maturity, while plant height could be directly evaluated. Nearly all the farmers perceived Kalinga III to be earlier than the local landrace, and Sathi-34-36 to be of a similar maturity. GR-3 was late maturing and this was probably the cause of its low yield in the marginal agricultural environment of the project. Farmers regarded Sathi-34-36 as taller than the local landrace and Kalinga III as being similar in height. The low yield of GR-3 could be explained further by its apparent short height, which is a disadvantage in weedy, drought-prone environments (Fig. 1).

Farmers were also asked to consider tiller number and panicle length. These traits are generally regarded as being less obvious than yield, time to mature and plant height, and possibly more difficult to assess. Farmers' perceptions were still clear and some cultivars were perceived by nearly all of the farmers to be worse than the local landrace. Again, GR-3 belonged to the group that was perceived to be markedly inferior to the local landrace, while Kalinga III was not perceived to be greatly different (Fig. 1).

The degree of agreement of farmers' perceptions of Kalinga III, the most preferred cultivar, was assessed across the six villages in the study. The differences were highly significant for grain yield, time to bloom and maturity, and in all cases Kalinga III was superior to the local landrace (Table 2). For yield, there was perfect agreement that Kalinga III were higher yielding in all villages in Madhya Pradesh and Gujarat (Fig. 2). However, perceptions that Kalinga III were higher yielding than the local landrace was far less marked in Rajasthan, but were always considered to be so by 50% or more of the farmers. The land is less sloping and more fertile in these Rajasthan villages, and so the advantage of Kalinga III, which escapes drought by its earliness, may have been reduced. Farmers' perceptions of time to bloom and time to maturity were in excellent agreement across the villages; in only one village, Simalda, was there a lack of consensus that Kalinga III was earlier to bloom and mature. In only two villages, both in Gujarat, did farmers perceive Kalinga III to be shorter than the local. It

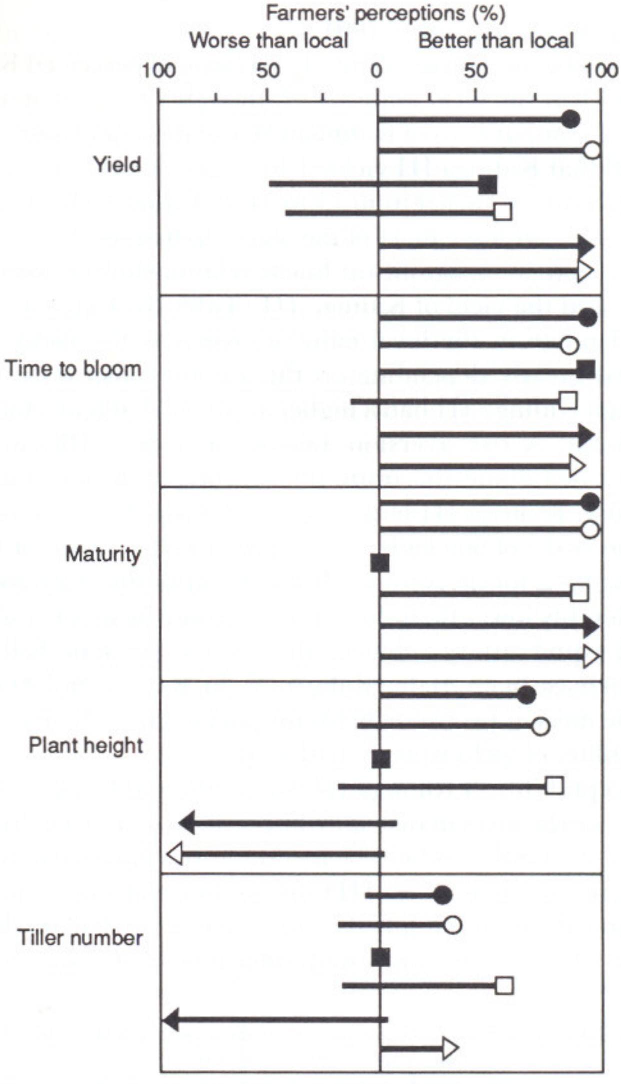

Fig. 2. Farmers' perceptions of the rice cultivar Kalinga III evaluated in six villages in the 1993 rainy season. Farmers' perceptions as to whether Kalinga III was better or worse than the local landrace are indicated by lines. The shorter the line the more farmers at that village responded that Kalinga III was the same as the local. —● Naganwat (Madhya Pradesh), —○ Palasiyapada (Madhya Pradesh), —■ Simalda (Rajasthan), —□ Kompura (Rajasthan). —▶ Sarjumi (Gujarat), —▷ Jaliyapada (Gujarat).

seemed likely that in these two villages the local landraces were taller. This meant that the overall difference between Kalinga III and the local landraces for plant height was not significant (Table 2). The overall difference for tiller number was also non-significant.

The focused group discussions showed that all groups perceived Kalinga III as having thin husks, grains that remained unbroken on dehulling, and grain that would fetch a higher market price than that of the local landraces.

Comparison of rice cultivars, rainy season 1994

In 1993, there was below average rainfall and farmers perceived Kalinga III to be higher yielding than the local landrace. Although 1994 was an exceptionally wet year, the paired comparisons of Kalinga III and the local landrace in terms of grain yield showed that Kalinga III yielded 46% more (Table 3). Across the four villages its yield advantage varied from 15 to 84%. Kalinga III yielded the same or more than the local landrace in 44 of the 58 comparisons (Fig. 3). In only one village, Sarjumi, was there a significant linear relationship between the yield of the local landrace and the yield of Kalinga III (Table 3). In most villages, there was a small yield range in the local cultivar, whereas the yield advantage of Kalinga III varied greatly depending on the abiotic constraints. Overall, this meant that although Kalinga III had a higher grain yield, it had a higher variance across farmers' fields. A risk aversion analysis was done (Barah *et al.*, 1981; Witcombe, 1988) to examine the trade off between increased grain yield and increased variability. Kalinga III had a mean grain yield of 1521 kg ha^{-1} with a standard deviation (s.d.) of 609 kg ha^{-1} compared with a mean of 1042 kg ha^{-1} with a s.d. of 428 kg ha^{-1} for the local landrace. This produced an iso-utility curve of only 0.3, considerably lower than the curve of 2.0 at which both cultivars would be considered of equal utility. Hence, the higher yield of Kalinga III far outweighed any increase in unpredictability of yield. Surveys in 1995 showed that farmers choose the most appropriate fields for particular cultivars carefully, and so the unpredictability of yield is likely to decline.

In 1994, the acceptability of Kalinga III was confirmed by sales of 3.5 t seed at unsubsidized prices to farmers in over 20 villages. Seed sales were limited by seed supply because the demand could not be met fully. In villages where no seed was sold large areas were sown to Kalinga III from seed saved by farmers. In addition, 415 kg seed of Sathi-34-36 was supplied to two villages, and again demand could not be satisfied. In 1995, 1.7 t seed was sold, since there were now large quantities

Table 3. *Comparison of Kalinga III with local landraces in farmer-managed paraticipatory research (FAMPAR) trials in four villages, rainy season* 1994

Village and state	Number of farmers	Yield (kg ha^{-1}) Kalinga III	Landraces	Yield of Kalinga III (% of local landraces)	*t*-value	r^2†
Kompura (Rajasthan)	31	1610	1036	155	4.4**	−0.05
Sadera (Rajasthan)	5	1711	927	184	2.53	0.22
Sarjumi (Gujarat)	14	1342	1161	115	1.83*	0.77*
Palasiyapada (Madhya Pradesh)	8	1373	929	147	1.28	−0.36
Overall	58	1521	1042	146	5.21**	0.13

†Coefficient of determination between yield of Kalinga III and local landraces across farmers; $p < 0.05$, $^{**}p < 0.01$.

of seed saved by farmers in the project villages. Two government agencies and 15 non-governmental organizations requested and were supplied with seed.

Comparison of chickpea cultivars, rabi 1992–93

Farmers were asked to compare the test cultivars with their local cultivar for many traits including vegetative growth, pest attack, market value, taste, seed size and colour, pod formation and crop maturity. Unlike the evaluation of the rice cultivars, preferences varied more across location, making it more difficult to assess farmers' preferences. This was because it was not possible to rank the importance of the different traits and the trade-off between them. However, from the farm walks, all farmers were aware of all the cultivars in the trials so they could make judgements on their overall preferences. Farmers were asked whether they would grow the cultivar they had tested for a second season, and whether they would purchase seed of a different test cultivar. The latter question revealed greater differences in farmers' preferences because there was now a choice amongst the four cultivars the farmers had not previously grown, and because purchasing seed was more onerous than replanting seed saved from the previous harvest (Table 4). If seed purchase was necessary, farmers preferred cultivars which demonstrated a good combination of earliness and high yield in the experimental plots (Fig. 4). ICCV 2 and ICCV 88202 had superior combinations of grain yield and earliness, followed by ICCV 10. The cultivars ICCV 37 and ICCC 4 were markedly inferior to the local cultivar, Dahod yellow, because they flowered later but failed to yield more. Farmers preferred early cultivars because grain prices were higher early in the season and because farmers perceived that

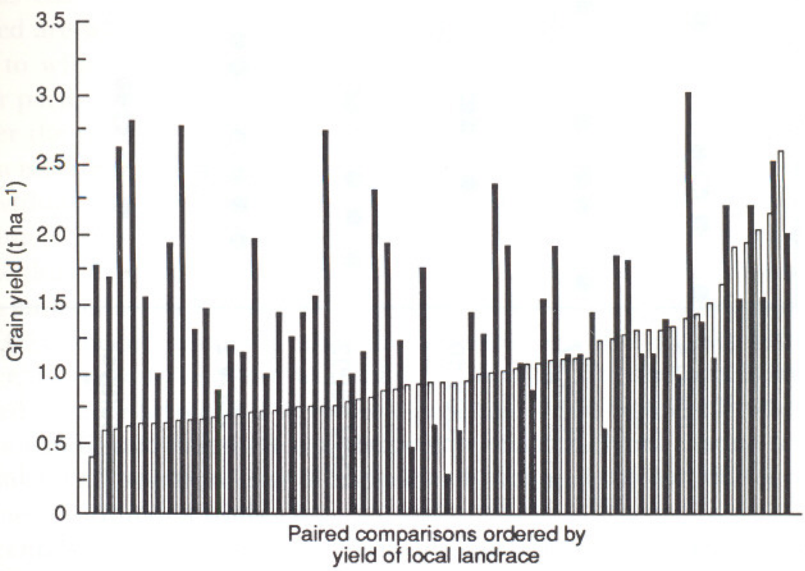

Fig. 3. Grain yield of the rice cultivar Kalinga III (■) and local landraces (□) in 58 paired comparisons in four villages, rainy season 1994.

Table 4. *Number of farmers participating in the 1992–93 farmer-managed participatory research (FAMPAR) trials who said they would grow one of the test chickpea cultivars in the following year*

Question	Sample size†	No. of farmers intending to grow specific chickpea cultivars				
		ICCC 4	ICCV 37	ICCV 10	ICCV 88202	ICCV 2
Will you regrow your 'own'‡ cultivar?	24	10	13	17	21	17
Will you grow another test cultivar?§	96	1	5	13	25	35
Total	120	11	18	30	46	52

†The number of farmers asked this question: for each chickpea cultivar, 24 farmers had grown it and 96 had not; ‡Own = test cultivar already grown by farmer; §Assuming the seed would be purchased at an unsubsidized price.

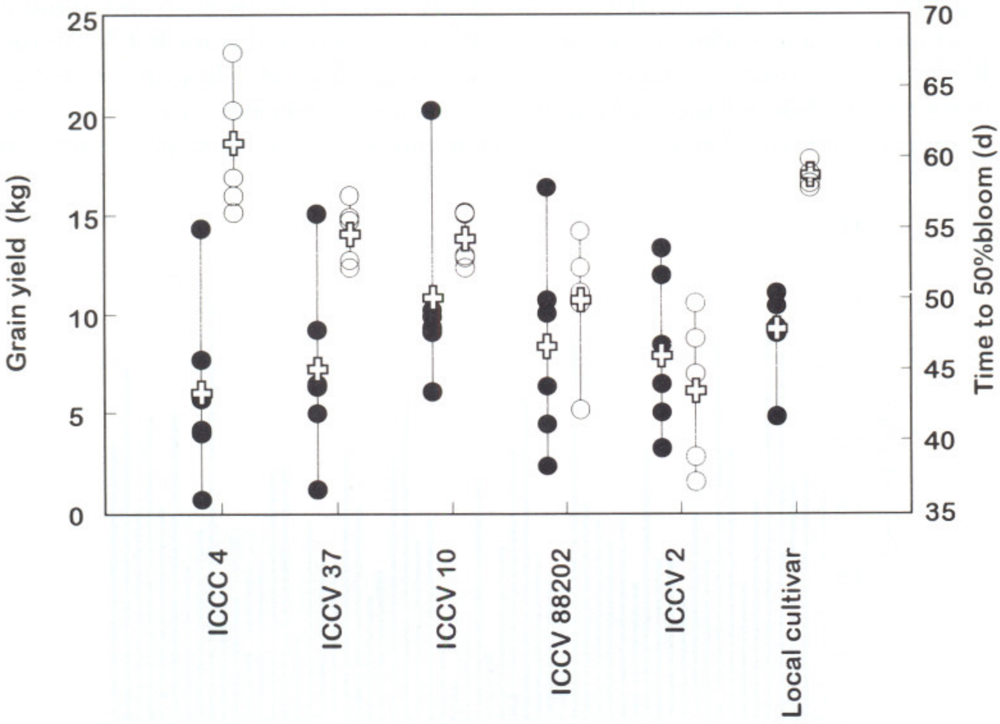

Fig. 4. Grain yield (●, kg plot^{-1} from 1 kg sown seed), in six villages, time to bloom (○) and in five of the six villages, and mean grain yield and time to bloom across villages (✚) of six chickpea cultivars, post-rainy season 1992–93.

earlier-maturing cultivars would escape drought caused by receding soil moisture. In 1995 the survey data were validated by farmers' subsequent behaviour. The three preferred cultivars had the highest percentage resowing, and were the only cultivars that farmers had given or sold to other farmers (Table 5).

The local cultivar showed less variation in grain yield across locations than the test varieties, and sometimes yielded more than them. However, farmers still preferred some of the test cultivars indicating that yield was not a paramount consideration for farmer selection.

The recommended cultivar, ICCC 4, was the least desirable and least preferred cultivar based on all criteria. No farmers chose to re-sow the late maturing ICCC 4 (Table 4), and when seed purchase was necessary, there was a 35-fold preference for ICCV 2, the preferred cultivar, compared with ICCC 4.

CONCLUSIONS

Decentralization of breeding programmes

A most important finding from the participatory varietal selection was that the domains of released cultivars are defined inadequately. Appropriate cultivars are not reaching farmers, and inappropriate cultivars are being recommended. The domain of Kalinga III is much greater than its area of release. It was the best performing cultivar in the participatory varietal selection trials, but was not released in any of the three states in which it was tested. Chickpea cultivars ICCV 2, ICCV 88202 and ICCV 37 were preferred by the farmers in the project area but are only recommended elsewhere. In contrast, the domains for which cultivars such as GR-3 and ICCC 4 (adapted to favourable environments) are recommended are whole states, so farmers are encouraged to grow them in marginal areas to which they are unadapted. Decentralized breeding, zonal trials and farmer participation would better define cultivar domains.

After the co-ordinated trials and national release, the first level of decentralization is state-level release. It allows plant breeders to target an agro-ecological zone within a state. For example, Dr Srinavas Rao, of the Central Rice Research Institute (CRRI), bred Kalinga III and obtained its release in Orissa using data from within-state, off-station trials (J. K. Roy, CRRI personal communication). In the national trials Kalinga III was rejected, having performed poorly in 1981 and 1982 in the preliminary trials of the All India Coordinated Rice Improvement Project. In these trials there were high levels of inputs, the test locations were situated in more favourable agro-ecological zones and high coefficients of determination indicated that there were large experimental errors. A state-level system is useful if it allows the release of a specifically adapted cultivar that is rejected by national co-ordinated trials.

Decentralization of breeding is desirable, but a potential disadvantage needs to be overcome. Although state-level breeding aids the production of cultivars adapted to specific within-state environments, it can also hinder the popularization of cultivars that originate outside. Many cultivars, such as Kalinga III,

Table 5. *Results of a survey in two villages in 1995 to determine resowing rates of chickpea cultivars first grown in* rabi *1993–94*

Chickpea cultivar	Sample size	Percentage resowing	Increase in area in project village (times)†	Number of new farmers given seed	Mean distance of new farmers from village (km)	Total seed given (kg)
ICCC 4	5	0	–	0	–	0
ICCV 37	4	50	2.4	0	–	0
ICCV 10	7	57	2.9	5	3.4	7
ICCV 88202	8	75	1.8	10	10.5	25
ICCV 2	8	75	2.8	4	5.6	3

†Ratio of seed sown in 1994–95 to seed sown in 1993–94. Area sown in 1994–95 excludes new farmers given seed in 1993–94 as data were unavailable.

ICCV 2 and ICCV 88202, are adapted to drought-stressed, infertile environments which occur in many states. Adapted cultivars need to be released in all such states but, instead of allowing this to be easy and rapid, states favour their own cultivars. A solution is to have trials of state-released cultivars in well-defined agro-ecological zones that will cross state boundaries. Such trials would be highly efficient:

- Allocating entries to them correctly would be helped by the extensive prior knowledge of the cultivars.
- The number of entries would be lower than in the coordinated trials, permitting the use of much larger plots.
- On-farm testing could be more easily incorporated.

With a system of zonal trials established, all released cultivars with appreciable adoption in any state would automatically be tested across states. Delays, such as the one of over ten years in testing Kalinga III in Rajasthan, Gujarat and western Madhya Pradesh after its release in Orissa, would be avoided. This delay occurred despite farmers' acceptance of Kalinga III in Bihar and eastern Madhya Pradesh without the support of official release.

Extension of approach to other ecosystems

Participatory varietal selection (PVS) is useful not only in marginal environments, but in high-potential irrigated ecosystems. Farmers adopt new cultivars slowly in these areas because they are exposed to a limited choice of cultivars and seed of new cultivars reaches farmers long after its release. For example, in areas of Gujarat where KRIBP is multiplying Kalinga III, only a few rice cultivars, GR-4 (released 1981), IR36 (1981) and GR-11 (1977), are grown. All farmers that multiplied Kalinga III retained seed for their own use and there are many other recent, high-yielding cultivars, widely adopted elsewhere, which farmers could be given for a PVS programme. In the more favourable areas of Rajasthan and Gujarat, where seed multiplication of the chickpea cultivars was undertaken, farmers retained seed sufficient to plant substantial areas.

Participatory data in release proposals

Farmers' perceptions based on focused group discussions and household level questionnaires varied little between the villages. This contrasted with the situation usually found in multi-locational on-station trials where the ranking of cultivars differs greatly between test locations. The quality of data on farmers' perceptions from well-conducted participatory trials is convincing (Table 2). It is more relevant than conventional multi-locational trial data, which is subject to much greater genotype–environment interactions, and which has yield as the overwhelmingly important selection criterion. For rice, reliable evaluations were made of traits such as grain breakage on dehusking, ease of dehusking, and market price, and these would not be measured in conventional trials. For chickpea, market price could be evaluated, and the trade-off between earliness and yield could be appreciated.

Unlike data from replicated yield trials, participatory data are not widely accepted by the scientific community. To provide data required by release committees, resources have to be spent on collecting yield data, the most important criterion for release, only to confirm the results obtained from farmers' perceptions. For chickpea, where the lack of increased yield was balanced against earliness and market price, even yield trials would not provide the necessary data for a release proposal. Participatory data need to be accepted more widely, particularly in the submission of release documents, and a participatory approach should be incorporated as an essential component of more breeding programmes.

Important features in the KRIBP approach

The approach for participatory varietal selection (PVS) used in KRIBP differed from Maurya and colleagues (1988) because most of the genotypes tested were released cultivars and not advanced lines. The released cultivars tested were not known in many parts of their area of adaptation because their domains were defined inadequately and they were not sufficiently popularized. The advantages of using released cultivars, and the reasons why such cultivars may include ones that are acceptable to resource-poor farmers have been reviewed (Witcombe *et al.*, 1996). The important features of the research were:

- Participatory rural appraisal to identify the major characteristics of local landraces. Cultivars obviously unacceptable for easily identified characters such as grain type, crop duration, and plant height could be avoided.
- Identification of suitable improved, released cultivars to provide a large 'basket of choices' (Chambers 1989) for farmers.
- Testing in the most appropriate environment—the farmers' fields with unchanged farmer-management.
- Farmers and their families assess all major parameters relevant to farmers, for example, taste, cooking quality, market value, and not just the limited set of characteristics measured in plant breeders' trials.
- Assessment of cultivar acceptability by measuring adoption by farmers.

These results confirmed those of Maurya and colleagues (1988), that farmers can obtain considerably higher yields by growing different cultivars without changes in management. It means that the promotion of a recommended 'package of practices' can be counter-productive, since it dissuades farmers averse to taking risks from trying new cultivars. It also persuades plant breeders to test cultivars under high levels of management that are atypical of resource-poor farmers' fields.

PVS was effective and reliable for identifying appropriate cultivars for resource-poor farmers. Partnerships between breeders and non-governmental organizations would be a most effective way of carrying out PVS programmes. Wider adoption by both governmental and non-governmental organizations of the farmer-participatory methodology advocated in this paper should result in rapid adoption of higher-yielding cultivars in marginal environments.

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